STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 4 -

Assist. Prof. Dr. Muzaffer BÖREKÇİ
Email: structuralanalysis1mb@gmail.com

3. Determinacy and Stability

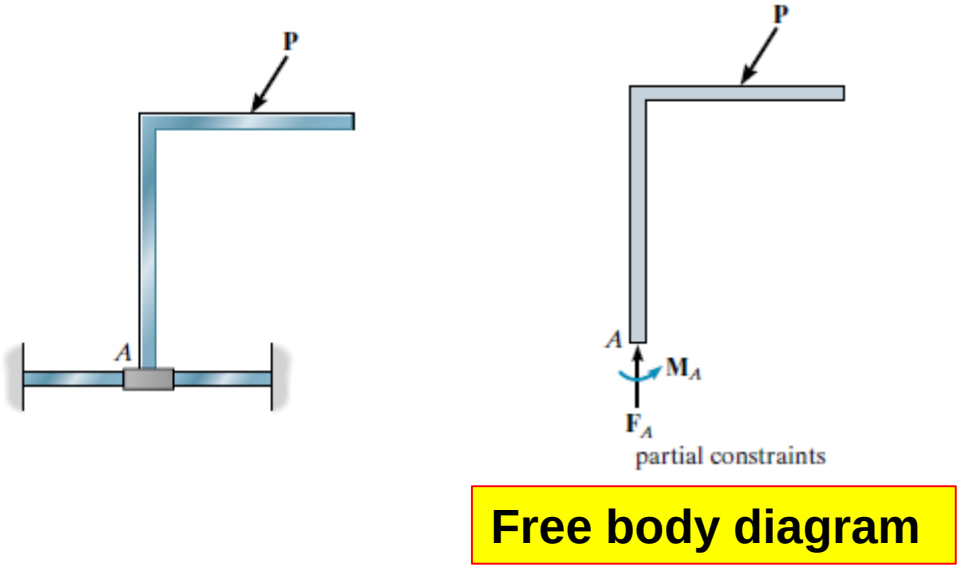
Stability

To ensure the equilibrium of a structure or its members, it is not only necessary to satisfy the equations of equilibrium, but the members must also be properly held or constrained by their supports regardless of how the structure is loaded. Two situations may occur where the conditions for proper constraint have not been met.

- Partial constraints
- Improper constriants

• Partial Constraints

Instability can occur if a structure or one of its members has *fewer* reactive forces than equations of equilibrium that must be satisfied. The structure then becomes only *partially constrained*.

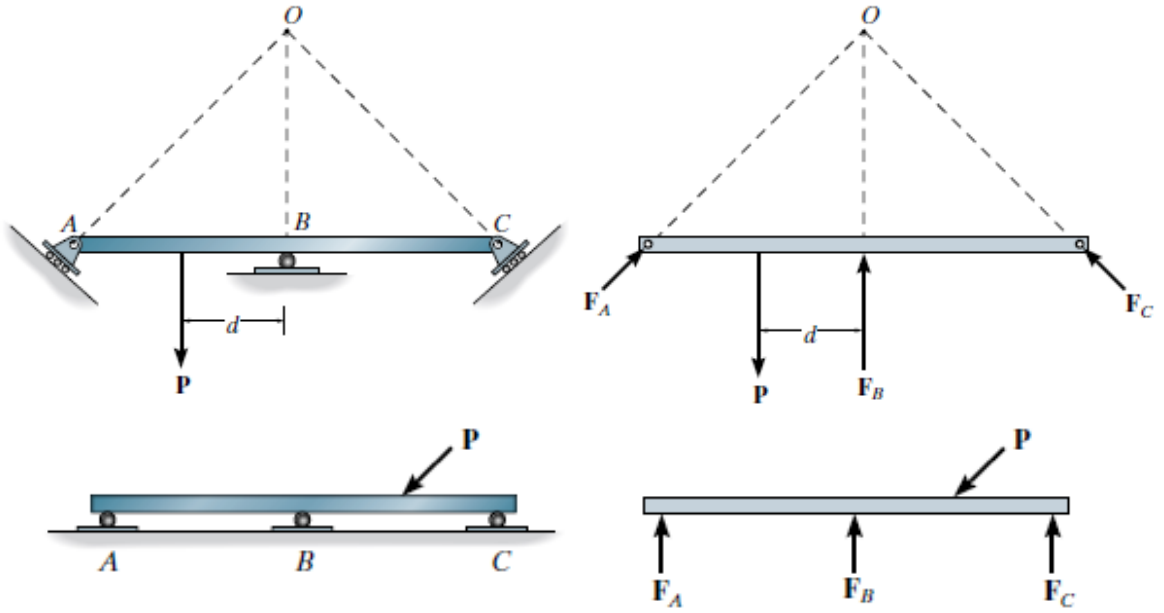


Free body diagram

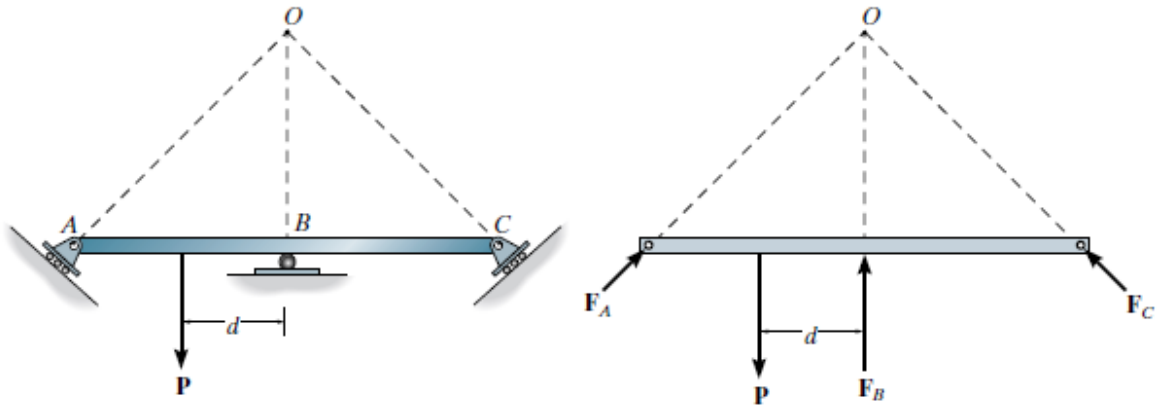
The equation $\sum P_x = 0$ will not be satisfied for the loading conditions, and therefore the member will be unstable.

- **Improper Constraints**

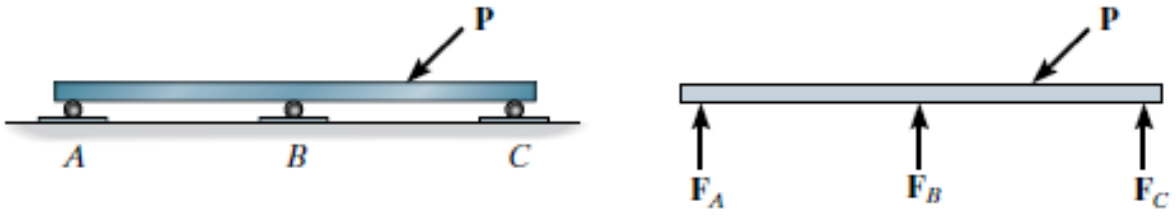
In some cases there may be as many unknown forces as there are equations of equilibrium; however, *instability* or movement of a structure or its members can develop because of *improper constraining* by the supports. This can occur if all the *support reactions are concurrent* at a point.



parallel reactions



An example of this is shown in the figure. From the free-body diagram of the beam it is seen that the summation of moments about point *O* will *not* be equal to zero; thus rotation about point *O* will take place.



parallel reactions

Another way in which improper constraining leads to instability occurs when the *reactive forces* are all *parallel*. An example of this case is shown in the figure. Here when an inclined force **P** is applied, the summation of forces in the horizontal direction will not equal zero

In general, a structure will be geometrically unstable

- if there are fewer reactive loads than equations of equilibrium
- although there are enough reactions, if the line of action of the reactive forces intersect at an common point or are parallel to one another

In all cases such types of structures must be avoided in practice.

Determinacy

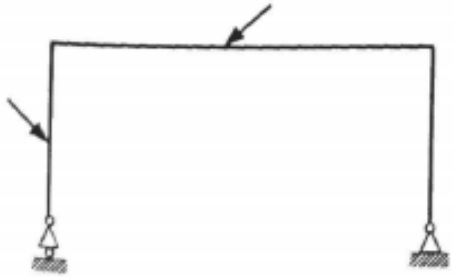
The equilibrium equations provide both the *necessary and sufficient* conditions for equilibrium.

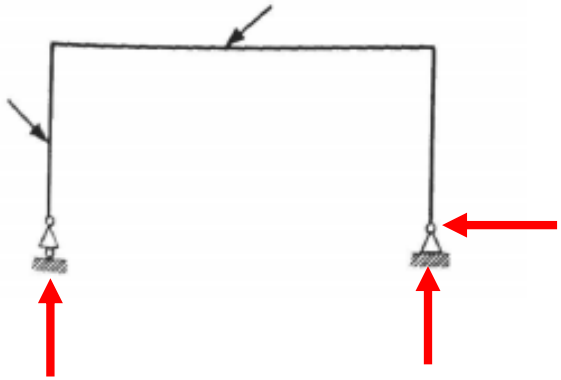
When all the forces in a structure can be determined strictly from these equations, the structure is referred to as *statically determinate*.

Structures having more unknown forces than available equilibrium equations are called *statically indeterminate*. If a structure is *statically indeterminate,* the additional equations needed to solve for the unknown reactions are obtained by relating the applied loads and reactions to the displacement or slope at different points on the structure.

This type of problem will be considered in Structural Analysis II

Statically determinate system

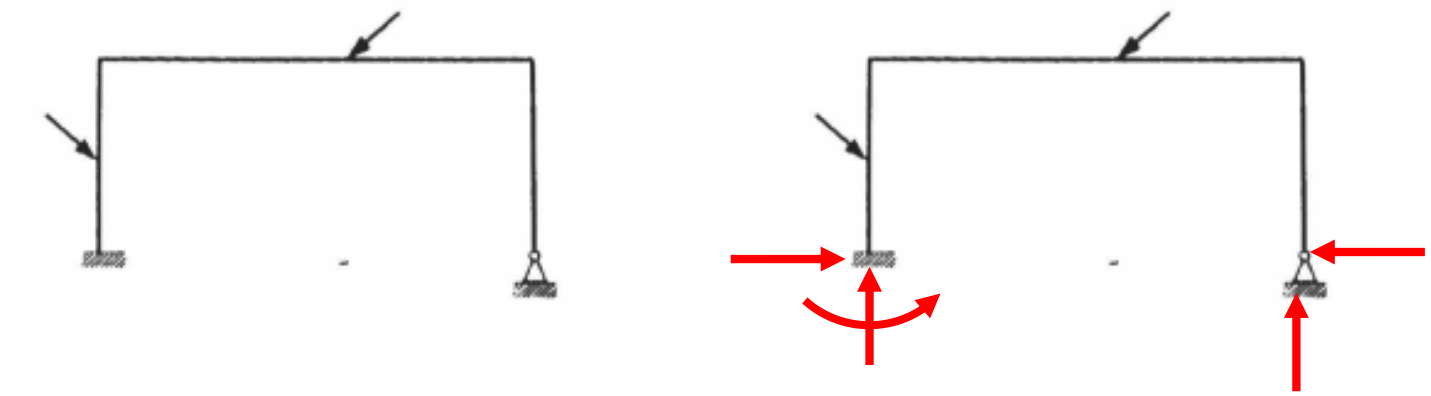




$n_r = 3$ (Number of reactions)

$n = 3 - 3 = 0$

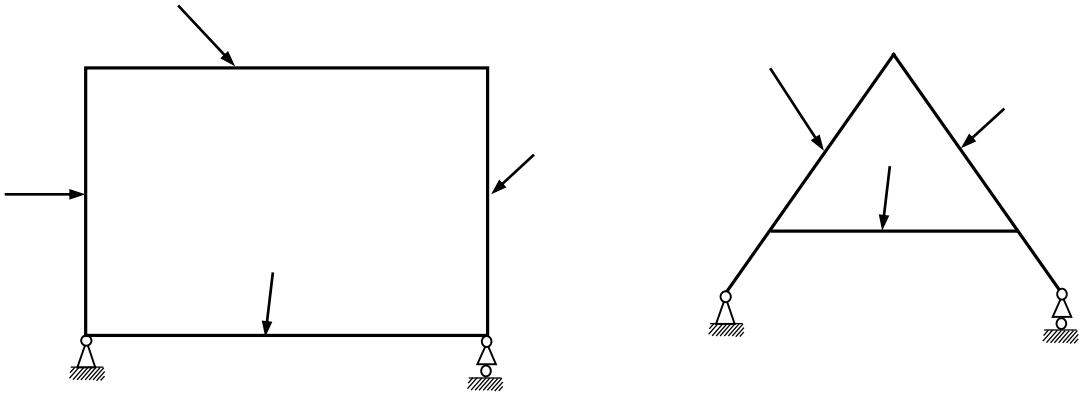
Statically indeterminate system



$n_r = 5$ (Number of reactions)

$n = 5 - 3 = 2$

Structures whose support reactions can be determined from the equations of equilibrium, but internal forces cannot be determined from the equations of equilibrium are referred to as *externally determinate, internally indeterminate*.



Hinge (Moment release) condition:

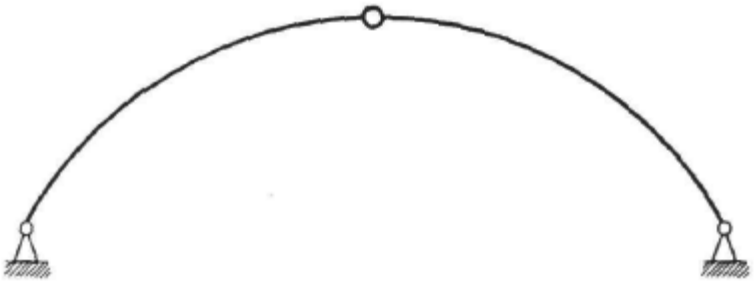
In civil engineering structures, the most common type of hinge is bending moment hinge.

M = 0 in the joint with hinge and the hinge reduces the degree on indeterminacy by 1 by providing an additional equation to the equations of equilibrium. In other words, since the total moment of the loads which place at any side of hinge (left or right) respect to hinge is zero, each hinges, exists in the system, adds an equation to equations of equilibrium.

Hinge (Moment release) condition:



An arc, statically indeterminate to the 1st degree

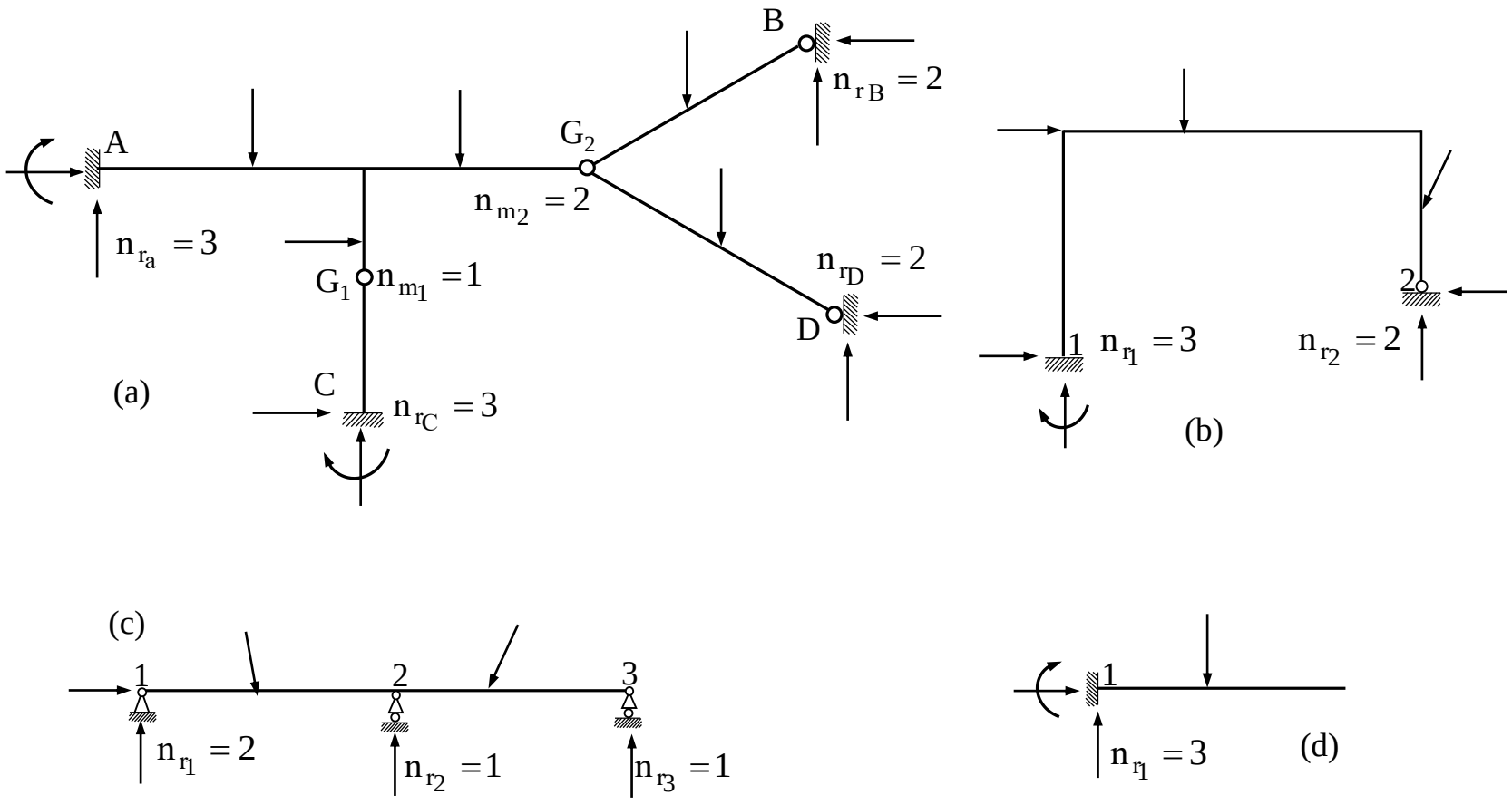


Statically determinate three-hinged arc

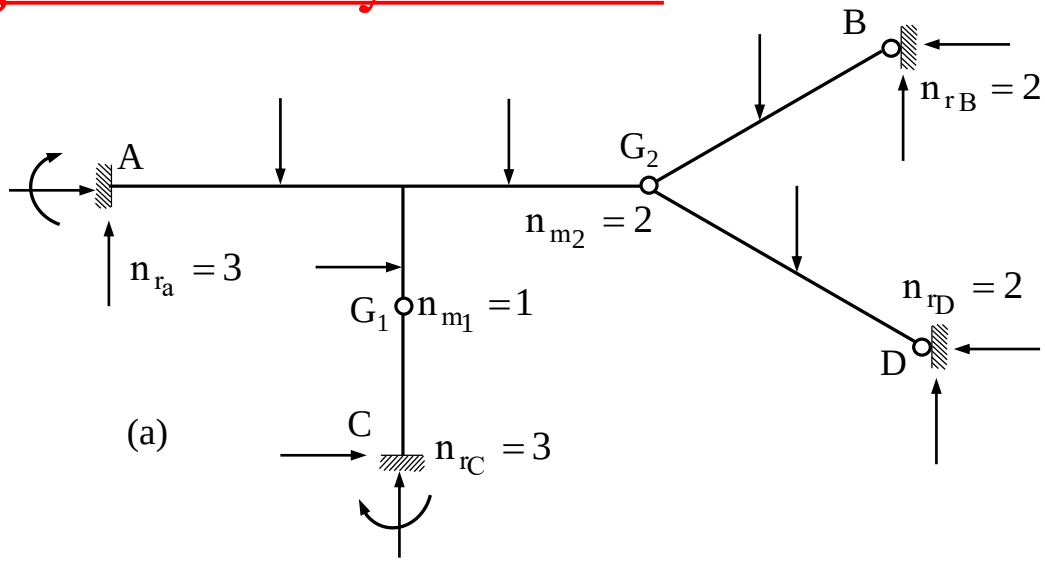
Degree of Indeterminacy

There are various methods in the estimation of the degree of indeterminacy of structures. Three of them are considered here.

1-Structures that can be separated into two parts with a single cut of any element



Structures that can be separated into two parts with a single cut of any element



- n_r : number of the unknown support reactions
- n_m : additional equations with hinges
- n_{el} : element number connected to hinge

$$n_m = n_{el} - 1$$

$$\mathbf{n = n_r - (n_m + 3)}$$

- n_r : number of the unknown support reactions
- n_m : additional equations with hinges
- n_{el} : element number connected to hinge

$$n_m = n_f - 1$$

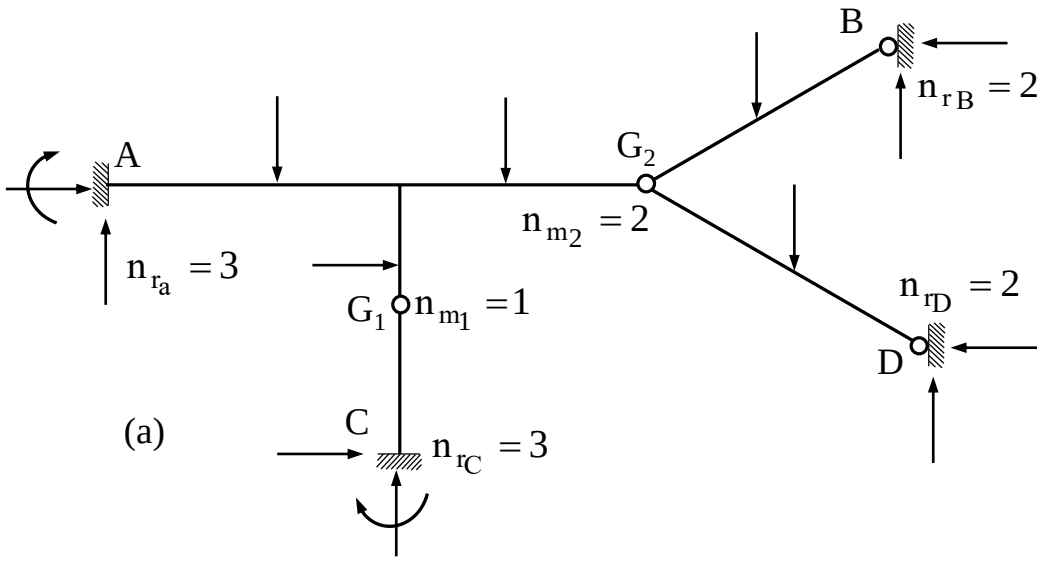
$$\mathbf{n = n_r - (n_m + 3)}$$

n = 0 (statically determinate)

n < 0 (unstable)

n > 0 (statically indeterminate to the n-th degree)

Example 13:



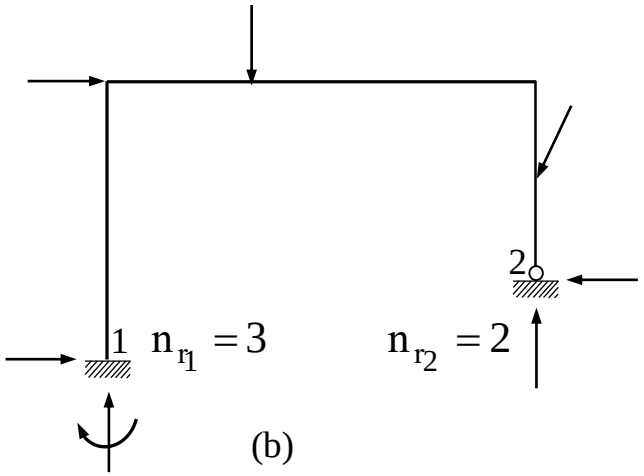
- $n_r = 10$
- $n_m = 3$

$n = 10 - (3 + 3)$

$n = 4$

Indeterminate to the 4th degree

Example 14:



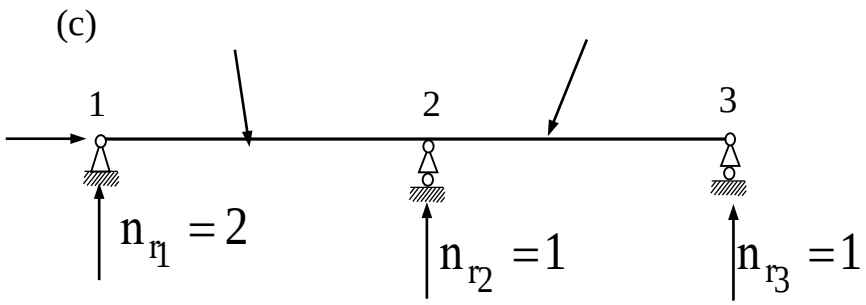
- $n_r = 5$
- $n_m = 0$

$n = 5 - (0 + 3)$

$n = 2$

Indeterminate to the 2nd degree

Example 15:



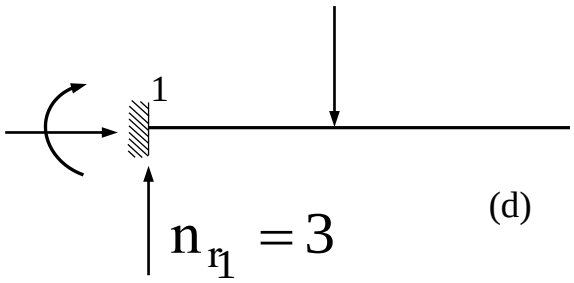
- $n_r = 4$
- $n_m = 0$

n = 4 – (0 + 3)

n = 1

Indeterminate to the 1st degree

Example 16:



- $n_r = 3$
- $n_m = 0$

n = 3 – (0 + 3)

n = 0

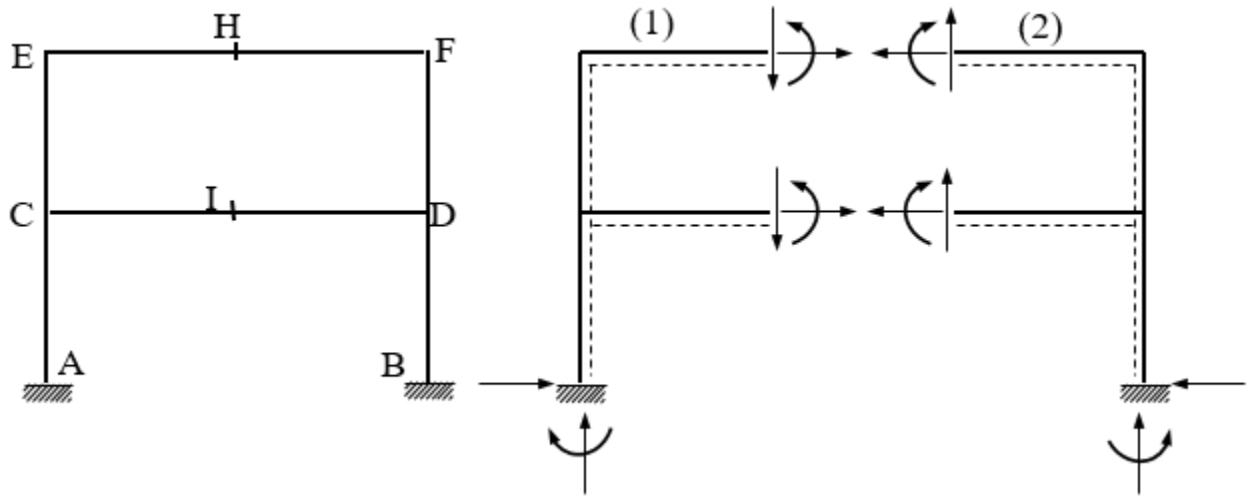
Statically determinate

2- Method of sections for all types of structures

- The structure is separated into simple parts that does not have any hinges and/or closed frames by cutting at proper sections.
- Each part should be in equilibrium under external loads at the cut section. Then three equilibrium equations can be written for each part. If there are n_s parts, $3xn_s$ equilibrium equations can be written.
- The total number of support reactions n_r is determined
- The number of unknown internal forces at the cut sections n_{ik} .
- The degree of indeterminacy is

$$\boldsymbol{n = n_{ik} + n_r - 3n_s}$$

Example 17: Determine the degree of indeterminacy.



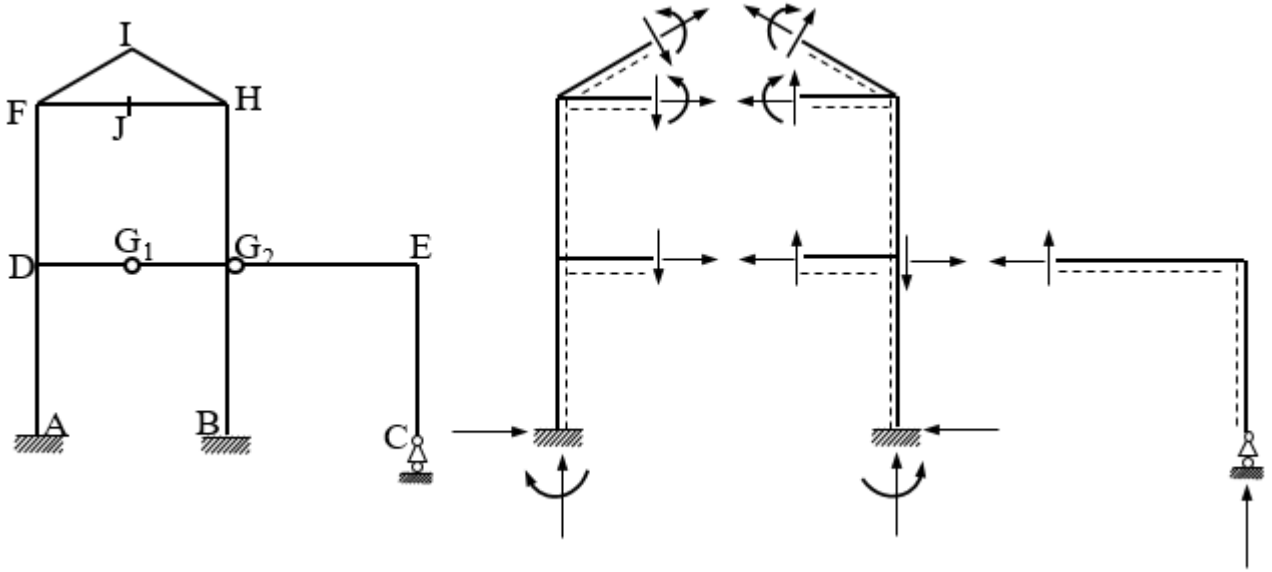
$n_s = 2$; $n_r = 6$; $n_{ik} = 6$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 6 + 6 – 3x2

n = 6

Example 18: Determine the degree of indeterminacy.



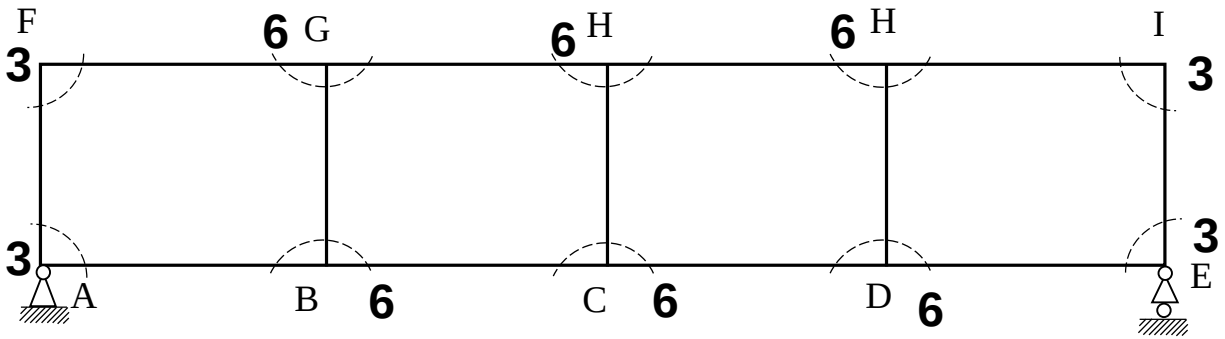
$n_s = 3$; $n_r = 7$; $n_{ik} = 10$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 10 + 7 – 3x3

n = 8

Example 19: Determine the degree of indeterminacy.



$n_s = 13$; $n_r = 3$; $n_{ik} = 48$

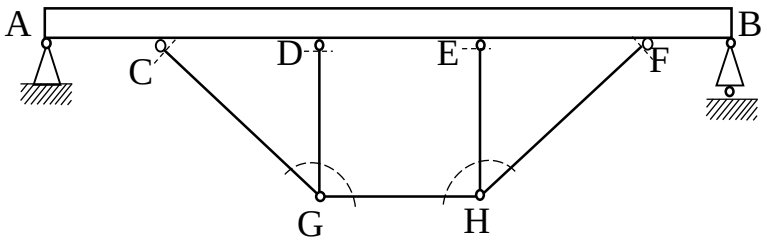
$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 48 + 3 – 3x13

n = 12

The structure is externally determinate since the support reactions can be determined by using the equations of equilibrium.

Example 20: Determine the degree of indeterminacy.



$n_s = 6$; $n_r = 3$; $n_{ik} = 16$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 16 + 3 – 3x6

n = 1

The structure is externally determinate since the support reactions can be determined by using the equations of equilibrium.

3- General methods

- A rigid joint permits the transfer of load and moment between the structural elements framing into the joint. According to the Bernoulli hypothesis, the rotation of the element framing into the same joint are equal to each other and equal to the rotation of joint itself.
- If a rigid frame composed of rigid joints is in equilibrium, then all structural elements and joints of the frame are also in equilibrium.
- Let «s» is the number of structural elements and j is the number of joints including the supports. Total number of equilibrium equations is:

$$n_{eq} = 3s + 3j$$

- There exist three internal forces at each end of structural member, thus six internal forces at each member. Then the total number of forces is

$$n_l = 6s + n_r$$

where n_r is the number of support reactions. Then, total number of unknown loads, that is, the degree of indeterminacy is

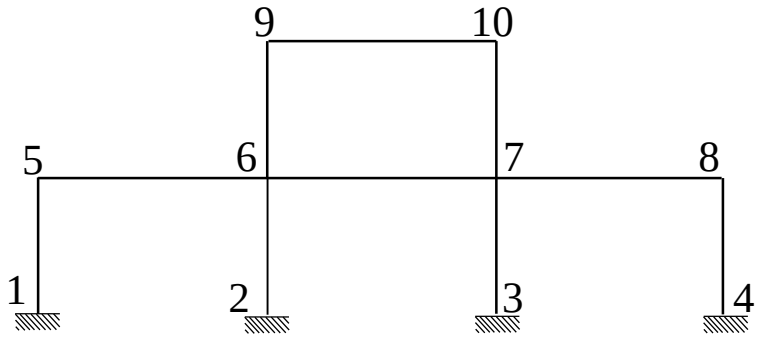
$$n = n_l - n_{eq} = 6s + n_r - 3s - 3j$$

$$\boxed{n = 3(s - j) + n_r}$$

If there exists any hinge in the structure, it decreases indeterminacy by 1, the number of equilibrium equations provided by hinges is introduced into the equation as

$$\boxed{n = 3(s - j) + n_r - n_m}$$

Example 21: Determine the degree of indeterminacy.



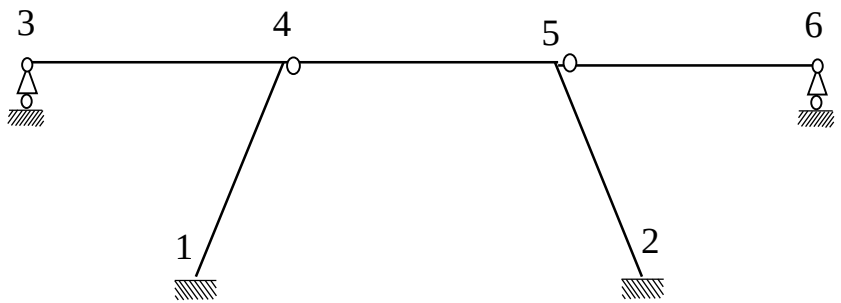
- The number of joints is j = 10
- The number of structural elements is s = 10
- The number of support reactions n_r = 4x3 = 12
- There is no hinge so n_m = 0

$$n = 3(s - j) + n_r - n_m$$

n = 3(10-10)+12-0

n = 12

Example 22: Determine the degree of indeterminacy.



- The number of joints is j = 6
- The number of structural elements is s = 5
- The number of support reactions n_r = 8
- Hinge condition n_m = 2

$$n = 3(s - j) + n_r - n_m$$

n = 3(5-6)+8-2

n = 3

3- Special case: Trusses

- Since the joints of trusses are pin joints, only two equilibrium equations can be written at each joints, so the number of equilibrium equations is

$$n_{eq} = 2j$$

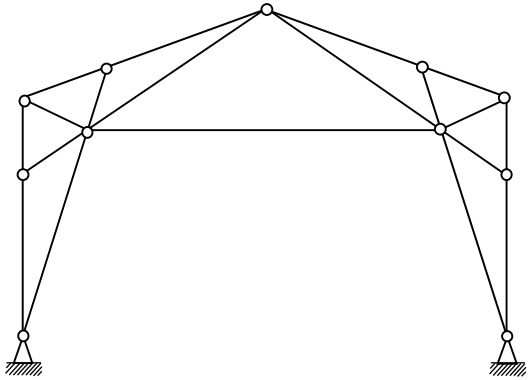
- Since bending moment and shear force in each element of a truss is zero, there is only one internal force (axial force). Then the total number of loads is

$$n_l = s + n_r$$

where n_r is the number of support reactions. Then, total number of unknown forces, that is, the degree of indeterminacy is

$$n = n_l - n_{eq} = (s - 2j) + n_r$$

Example 23: Determine the degree of indeterminacy.



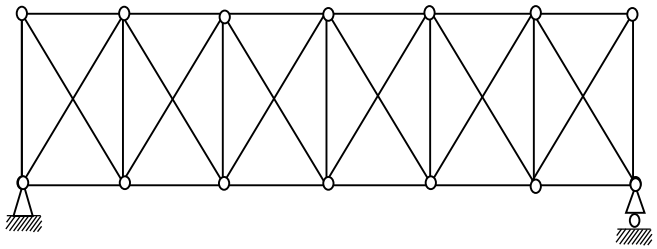
- The number of joints is j = 11
- The number of structural elements is s = 19
- The number of support reactions n_r = 4

$$n = (s - 2j) + n_r$$

n = (19-2x11)+4

n = 1

Example 24: Determine the degree of indeterminacy.



- The number of joints is j = 14
- The number of structural elements is s = 31
- The number of support reactions n_r = 3

$$n = (s - 2j) + n_r$$

n = (31- 2x14)+4

n = 6

Example 25: Determine the degree of indeterminacy.



First method: Since the system can be separated into two part

- n_r = 3 (number of the unknown support reactions)
- n_m = 0 (additional equations with hinges)

$$n = n_r - (n_m + 3)$$

$$n = 3-(0+3)$$

$$n = 0$$

Statically determinate

Example 25: Determine the degree of indeterminacy.



Second method:

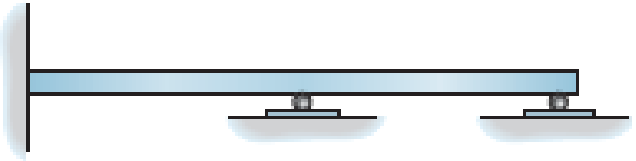
- The number of joints is j = 2
- The number of structural elements is s = 1
- The number of support reactions n_r = 3
- There is no hinge so n_m = 0

$$n = 3(s - j) + n_r - n_m$$

n = 3(1 - 2) + 3 - 0

n = 0

Example 26: Determine the degree of indeterminacy.



First method: Since the system can be separated into two part

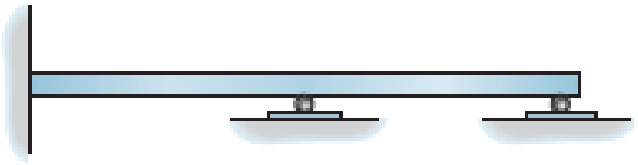
- n_r = 5 (number of the unknown support reactions)
- n_m = 0 (additional equations with hinges)

$$n = n_r - (n_m + 3)$$

$$n = 5 - (0 + 3)$$

$$n = 2$$

Example 26: Determine the degree of indeterminacy.



Second method:

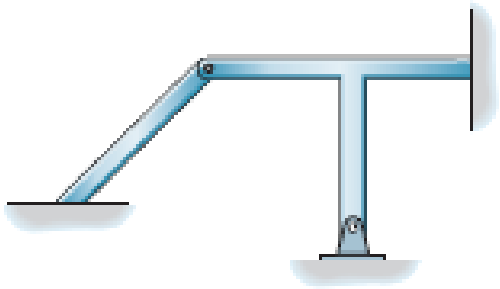
- The number of joints is j = 3
- The number of structural elements is s = 2
- The number of support reactions n_r = 5
- There is no hinge so n_m = 0

$$n = 3(s - j) + n_r - n_m$$

n = 3(2 - 3) + 5 - 0

n = 2

Example 27: Determine the degree of indeterminacy.



First method: Since the system can be separated into two part

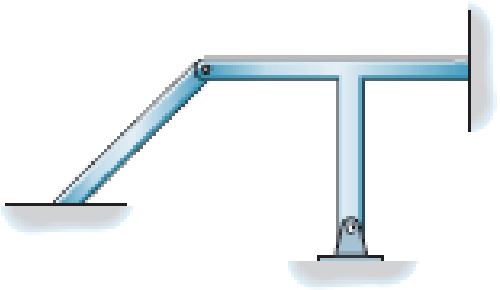
- n_r = 8 (number of the unknown support reactions)
- n_m = 1 (additional equations with hinges)

$$n = n_r - (n_m + 3)$$

$$n = 8 - (1 + 3)$$

$$n = 4$$

Example 27: Determine the degree of indeterminacy.



Second method:

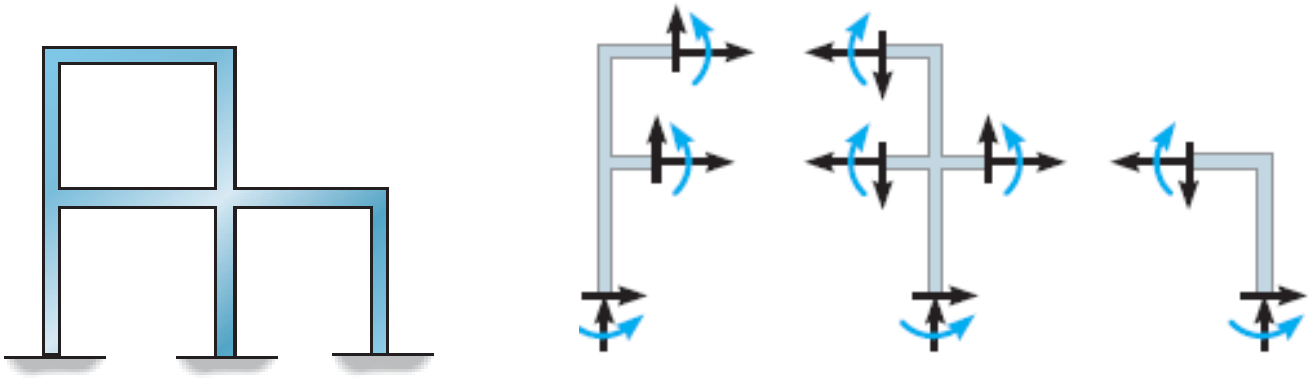
- The number of joints is j = 5
- The number of structural elements is s = 4
- The number of support reactions n_r = 8
- Hinge condition n_m = 1

$$n = 3(s - j) + n_r - n_m$$

n = 3(4 - 5) + 8 - 1

n = 4

Example 28: Determine the degree of indeterminacy.



First method: Since the system cannot be separated into two parts

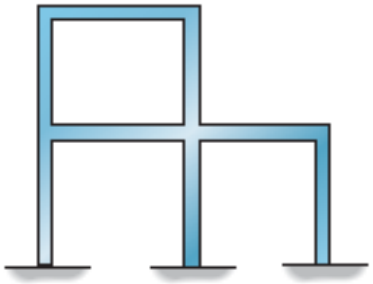
$$n_s = 3 \quad ; \quad n_r = 9 \quad ; \quad n_{ik} = 9$$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 9 + 9 – 3x3

n = 9

Example 28: Determine the degree of indeterminacy.



Second method:

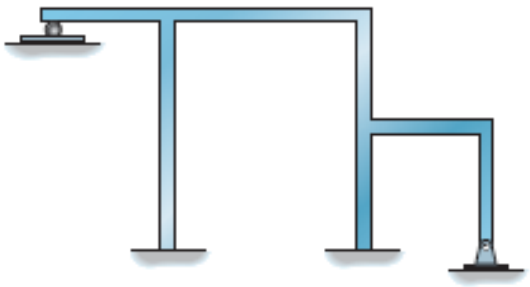
- The number of joints is j = 8
- The number of structural elements is s = 8
- The number of support reactions n_r = 9
- Hinge condition n_m = 0

$$n = 3(s - j) + n_r - n_m$$

n = 3(8 - 8) + 9 - 0

n = 9

Example 29: Determine the degree of indeterminacy.



First method: Since the system can be separated into two parts

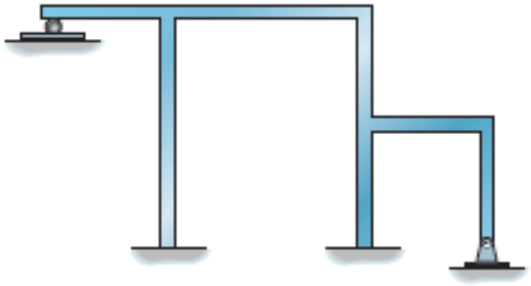
- n_r = 9 (number of the unknown support reactions)
- n_m = 0 (additional equations with hinges)

$$n = n_r - (n_m + 3)$$

$$n = 9 - (0 + 3)$$

$$n = 6$$

Example 29: Determine the degree of indeterminacy.



Second method:

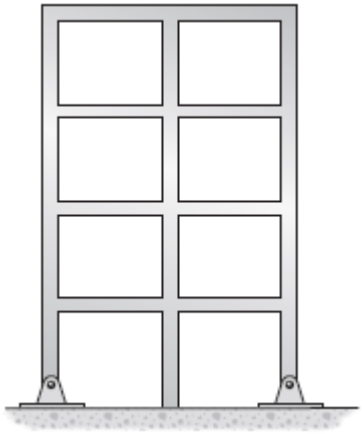
- The number of joints is j = 8
- The number of structural elements is s = 7
- The number of support reactions n_r = 9
- Hinge condition n_m = 0

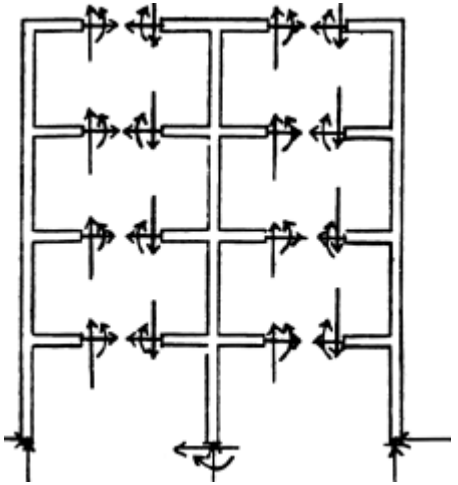
$$n = 3(s - j) + n_r - n_m$$

n = 3(7 - 8) + 9 - 0

n = 6

Example 30: Determine the degree of indeterminacy.





First method: Since the system cannot be separated into two part

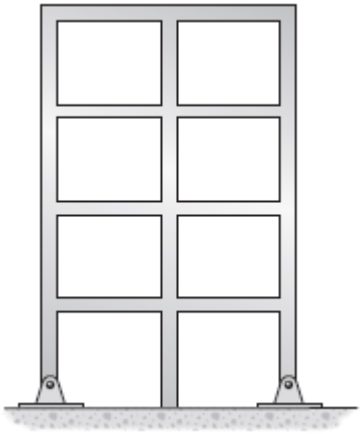
$$n_s = 3 \quad ; \quad n_r = 7 \quad ; \quad n_{ik} = 24$$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 24 + 7 – 3x3

n = 22

Example 30: Determine the degree of indeterminacy.



Second method:

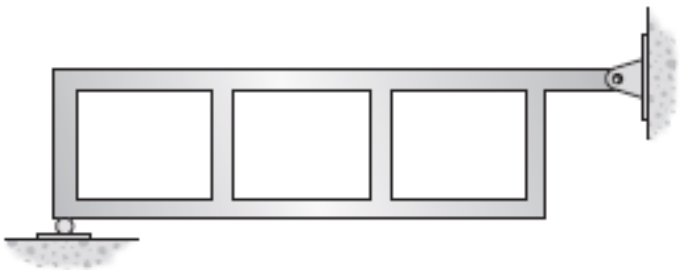
- The number of joints is j = 15
- The number of structural elements is s = 20
- The number of support reactions n_r = 7
- Hinge condition n_m = 0

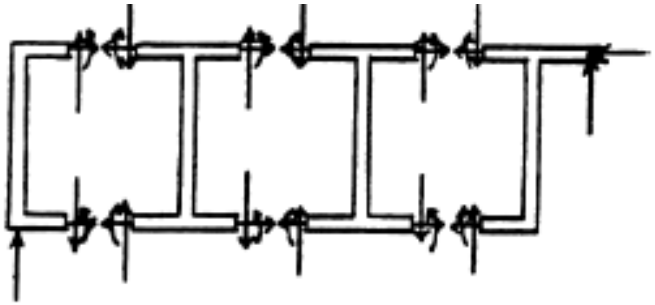
$$n = 3(s - j) + n_r - n_m$$

n = 3(20 - 15) + 7 - 0

n = 22

Example 31: Determine the degree of indeterminacy.





First method: Since the system cannot be separated into two part

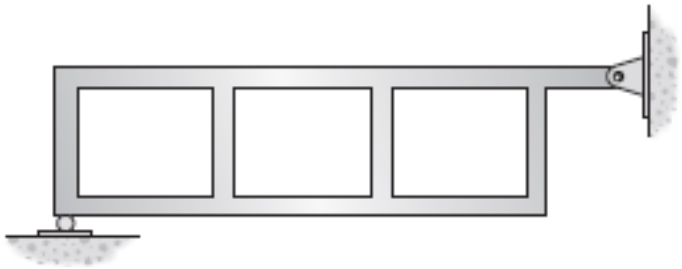
$$n_s = 4 \quad ; \quad n_r = 3 \quad ; \quad n_{ik} = 18$$

$$\mathbf{n = n_{ik} + n_r - 3n_s}$$

n = 18 + 3 – 3x12

n = 9

Example 31: Determine the degree of indeterminacy.



Second method:

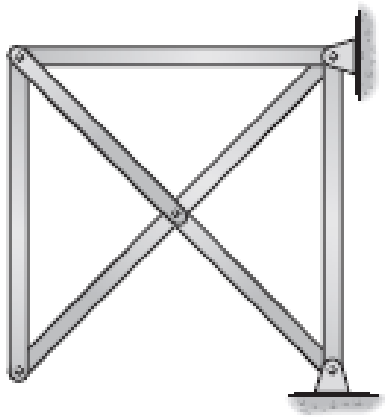
- The number of joints is j = 9
- The number of structural elements is s = 11
- The number of support reactions n_r = 3
- Hinge condition n_m = 0

$$n = 3(s - j) + n_r - n_m$$

n = 3(11 - 9) + 3 - 0

n = 9

Example 32: Determine the degree of indeterminacy.



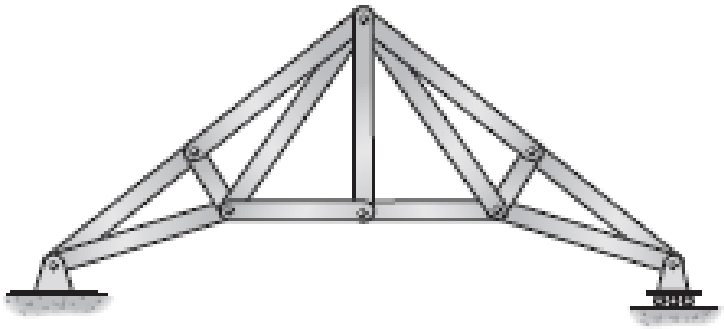
- The number of joints is j = 5
- The number of structural elements is s = 7
- The number of support reactions n_r = 4

$$n = (s - 2j) + n_r$$

n = (7 - 2x5)+4

n = 1

Example 33: Determine the degree of indeterminacy of the truss.



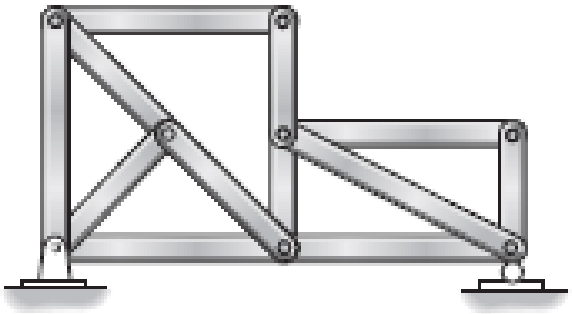
- The number of joints is j = 8
- The number of structural elements is s = 11
- The number of support reactions n_r = 3

$$n = (s - 2j) + n_r$$

n = (13 - 2x8)+3

n = 0

Example 34: Determine the degree of indeterminacy of the truss.



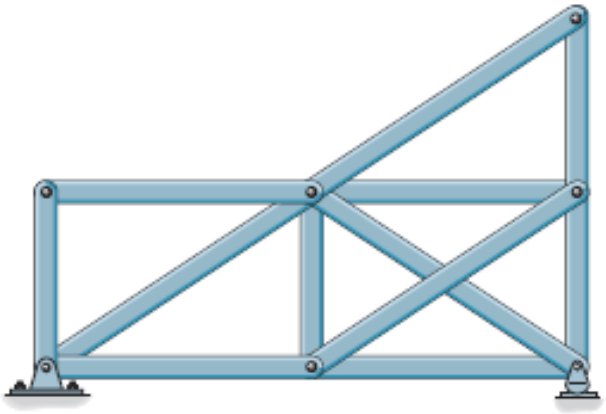
- The number of joints is j = 8
- The number of structural elements is s = 12
- The number of support reactions n_r = 3

$$n = (s - 2j) + n_r$$

n = 12 + 3 - 2x8

n = -1 (Unstable)

Example 35: Determine the degree of indeterminacy of the truss.



- The number of joints is j = 7
- The number of structural elements is s = 12
- The number of support reactions n_r = 3

$$n = (s - 2j) + n_r$$

n = 12 + 3 - 2x7

n = 1